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Microsoft Hololens

Report of Evidence of Use

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Patent Number: US8902225B2

Claim Number: (1)

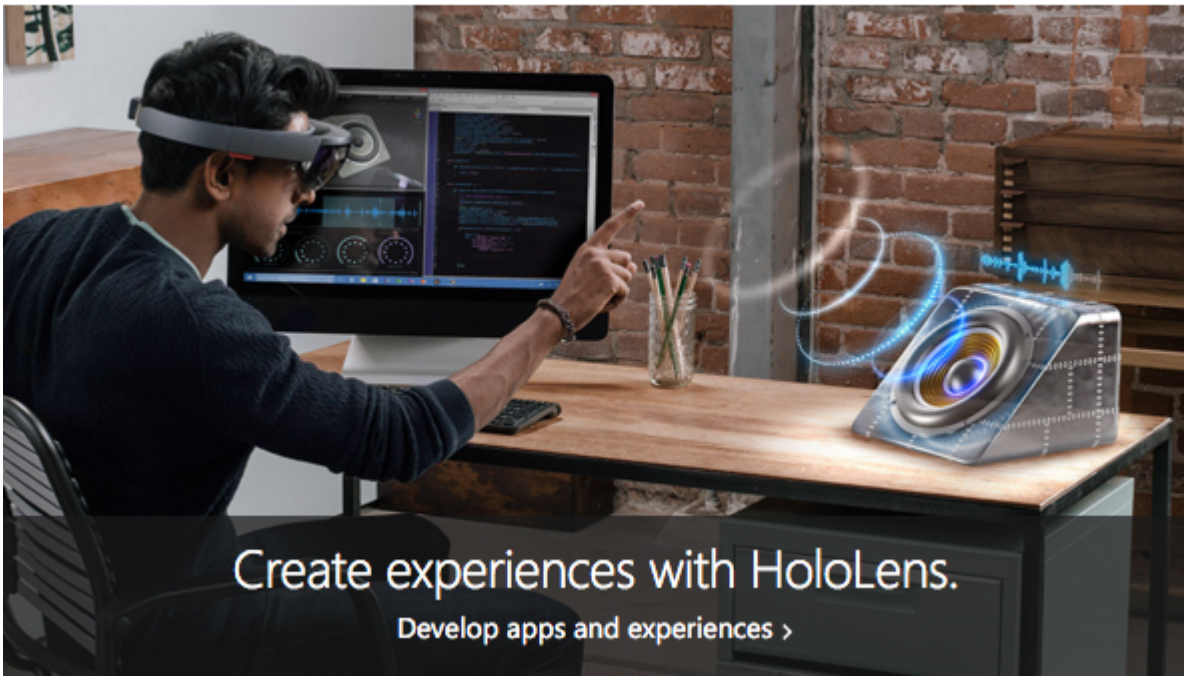
What is claimed is: 1. A user interface method, the method comprising: obtaining a projector display of an initial image; using one or more sensors coupled to a computer processor, sensing a user interaction with the displayed initial image, the user interaction being in a three dimensional coordinate system, and said sensing being based on quadrilateral angle navigation that determines a sensing point position within the three dimensional coordinate system, wherein an original image is displayed from a certain point in space preconfigured to match specific points in the quadrilateral angle navigation, and the certain point in space is matched to space location of the user interaction to determine the sensing point, said sensing resulting in each sensor providing coordinate information to the computer processor enabling the computer processor to correlate with the original image in the three dimensional coordinate system; in the computer processor, correlating the sensed user interaction with the three dimensional coordinate system and responsively manipulating the initial image resulting in an updated image based on the correlated user interaction; and from the computer processor through a projector, projecting the resulting updated image, the updated image being equivalent to the initial image manipulated by a distortion in three dimensional space.

Element 1:

What is claimed is: 1. A user interface method, the method comprising:

The Microsoft Hololens creates a immersive user interface system via its Hololens headset.

Supporting Evidence:



Element 2:

obtaining a projector display of an initial image;

The Hololens projector system displays an initial image, as displayed in the attached image from Microsoft's Hololens website..

Supporting Evidence:



Element 3:

using one or more sensors coupled to a computer processor,

The Hololens system has sensors coupled to a processor, as displayed in the attached image.

Supporting Evidence:



Element 4:

sensing a user interaction with the displayed initial image,

The Hololens senses the user interaction with the initial image.

Supporting Evidence:

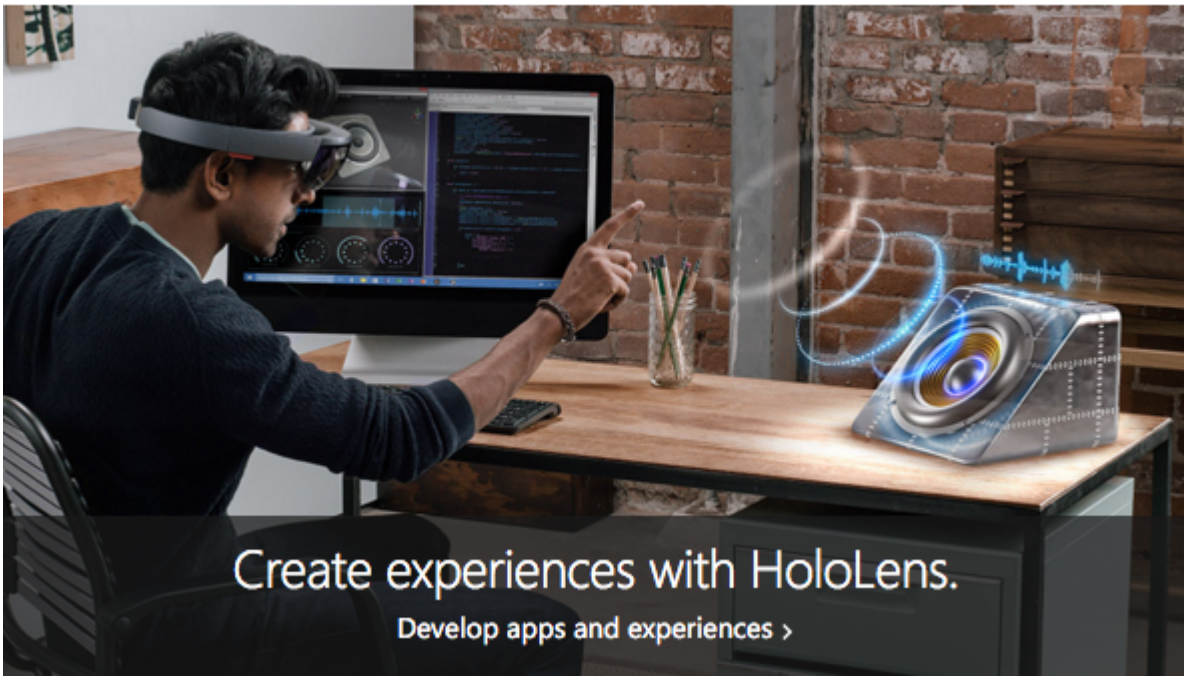


Element 5:

the user interaction being in a three dimensional coordinate system,

The user of the Hololens is reaching into space and this user interaction is taking place within a 3-dimensional coordinate system.

Supporting Evidence:



Element 6:

and said sensing being based on quadrilateral angle navigation

The Hololens has four sensors to perform object sensing quadrilateral angle navigation as well as other sensors for audio etc.

Supporting Evidence:



Engadget says "*There are at least four cameras or sensors on the front of the HoloLens prototype.*" Mashable writes "*Based on a quick look at the headset, it appears to have four front-facing cameras that could be used to detect the positions of the user's hands as she interacts with holographic objects.*"

Element 7:

that determines a sensing point position within the three dimensional coordinate system,

The Hololens determines the sensing point position of a users interactions with the projected images.

Supporting Evidence:



Element 8:

wherein an original image is displayed from a certain point in space

Here is an original image from the Hololens system being displayed in a point of space as experienced by the user.

Supporting Evidence:



Element 9:

preconfigured to match specific points in the quadrilateral angle navigation,

The Hololens 'pins' images to locations within the specific points which have been preconfigured by the Hololens system and are navigable by the user via the Hololens quadrilateral sensor systems.

Supporting Evidence:



Your world as a canvas.

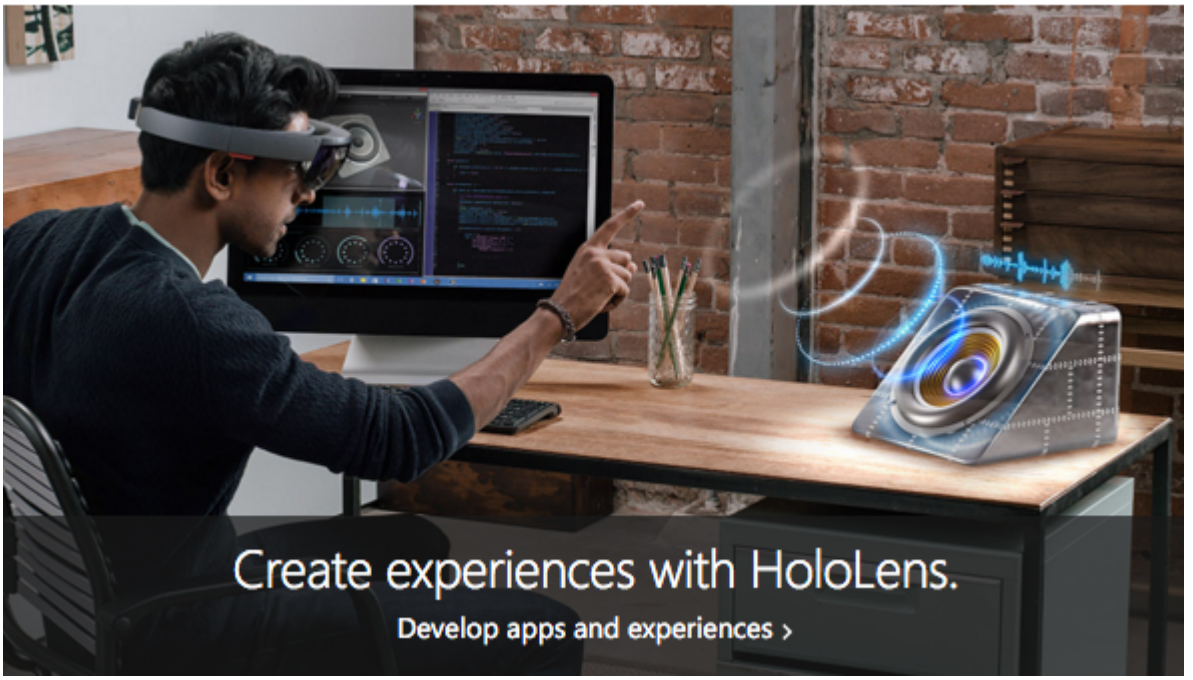
Microsoft HoloLens intelligently maps your room, mixing holograms with the environment around you. Pin holograms in physical locations as easily as you would place a physical object in a room. Interact with holograms and everyday objects together.

Element 10:

and the certain point in space is matched to space location of the user interaction to determine the sensing point,

Here is an image of a Hololens user reaching into space to a certain point, that point in space being matched to the user interaction, with the Hololens detrmining the sensing point

Supporting Evidence:

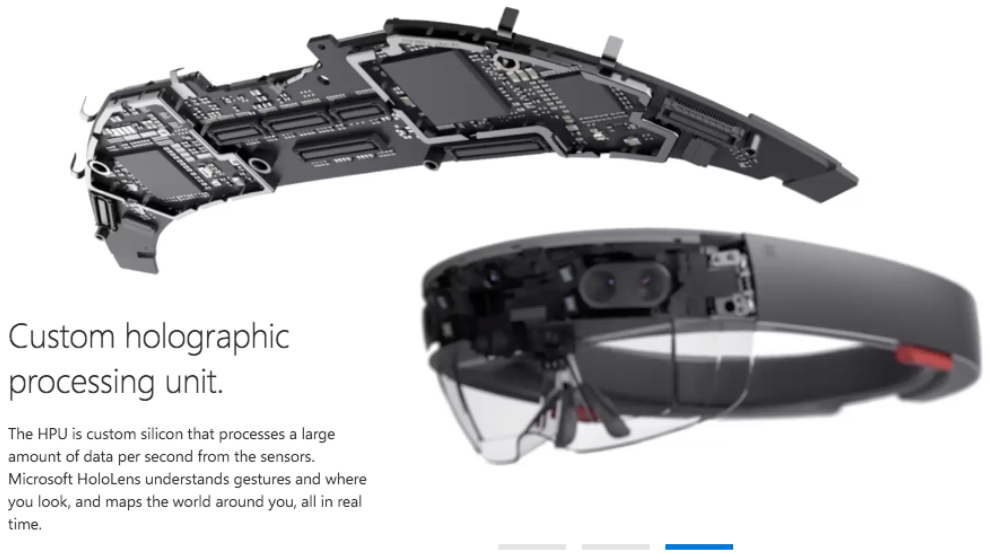


Element 11:

said sensing resulting in each sensor providing coordinate information to the computer processor

The attached image from Microsoft Hololens details how the Hololens system provides information from each system sensor, providing that coordinate and other sensor data to the Hololens processor.

Supporting Evidence:

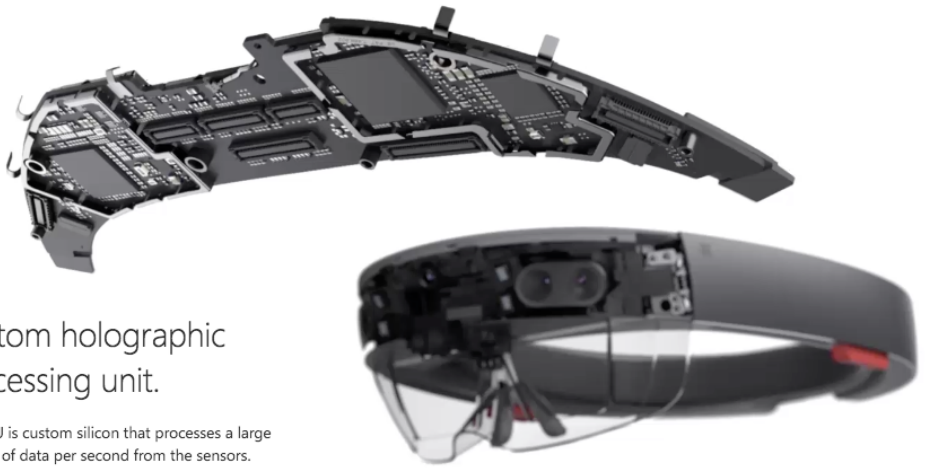


Element 12:

enabling the computer processor to correlate with the original image in the three dimensional coordinate system;

The Hololens system then correlates your data including the original image and 'maps' the data within this three dimensional coordinate system.

Supporting Evidence:



Custom holographic processing unit.

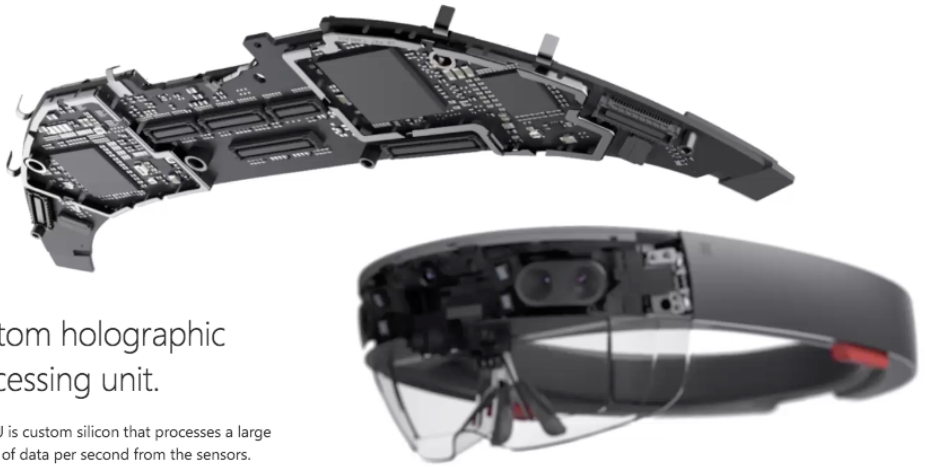
The HPU is custom silicon that processes a large amount of data per second from the sensors. Microsoft HoloLens understands gestures and where you look, and maps the world around you, all in real time.

Element 13:

in the computer processor, correlating the sensed user interaction with the three dimensional coordinate system

The Hololens system 'understands your gestures' via the system sensor array and correlates them to the world around you.

Supporting Evidence:



Custom holographic processing unit.

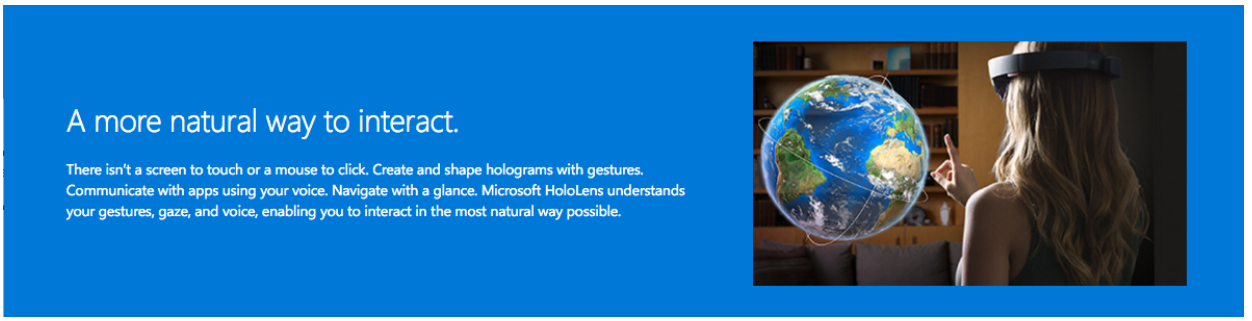
The HPU is custom silicon that processes a large amount of data per second from the sensors. Microsoft HoloLens understands gestures and where you look, and maps the world around you, all in real time.

Element 14:

and responsively

The system Hololens 'understands' responds to user interaction toward an image.

Supporting Evidence:

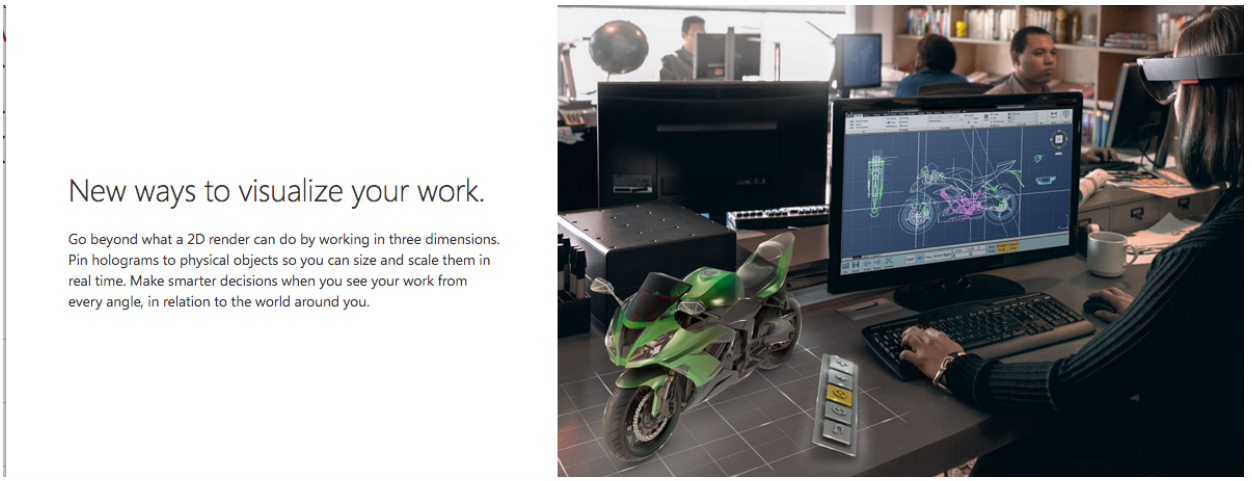


Element 15:

manipulating the initial image

At the core of the Hololens system functionality is the users ability to manipulate the 3D image. The attached image refers to image manipulation in the form of size and scale.

Supporting Evidence:



Element 16:

resulting in an updated image

The Hololens presents a new updated image via resizing or scaling, as well as shaping.

Supporting Evidence:



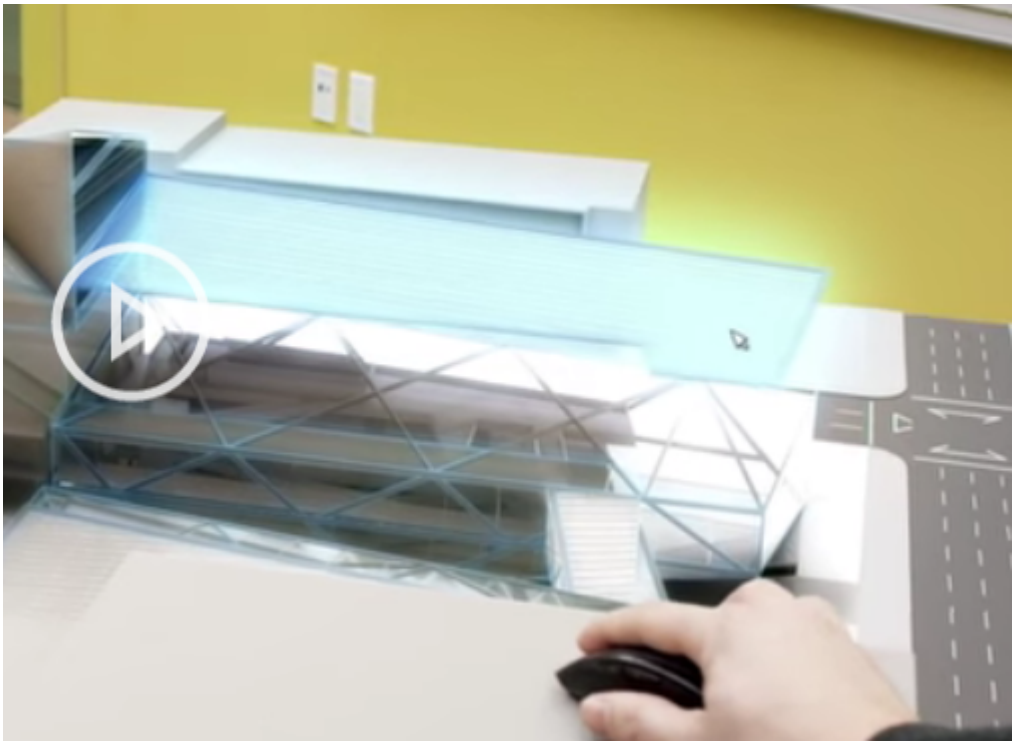
Element 17:

based on the correlated user interaction;

Here is an example of correlated user interaction using Hololens via Microsofy partner Trimble as the user raises the roof/floor of a building in 3D utilizing a mouse. The image responds to the correlated user interaction by raising the roof.

https://www.microsoft.com/microsoft-hololens/en-us/commercial

Supporting Evidence:

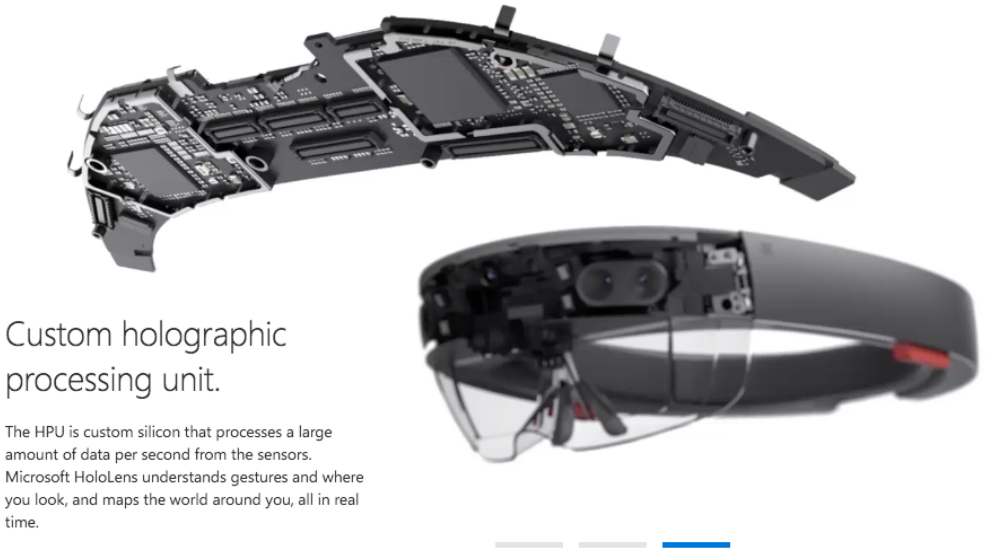


Element 18:

and from the computer processor through a projector,

The Hololens processor delivers instructions through the projector to create the user interaction.

Supporting Evidence:



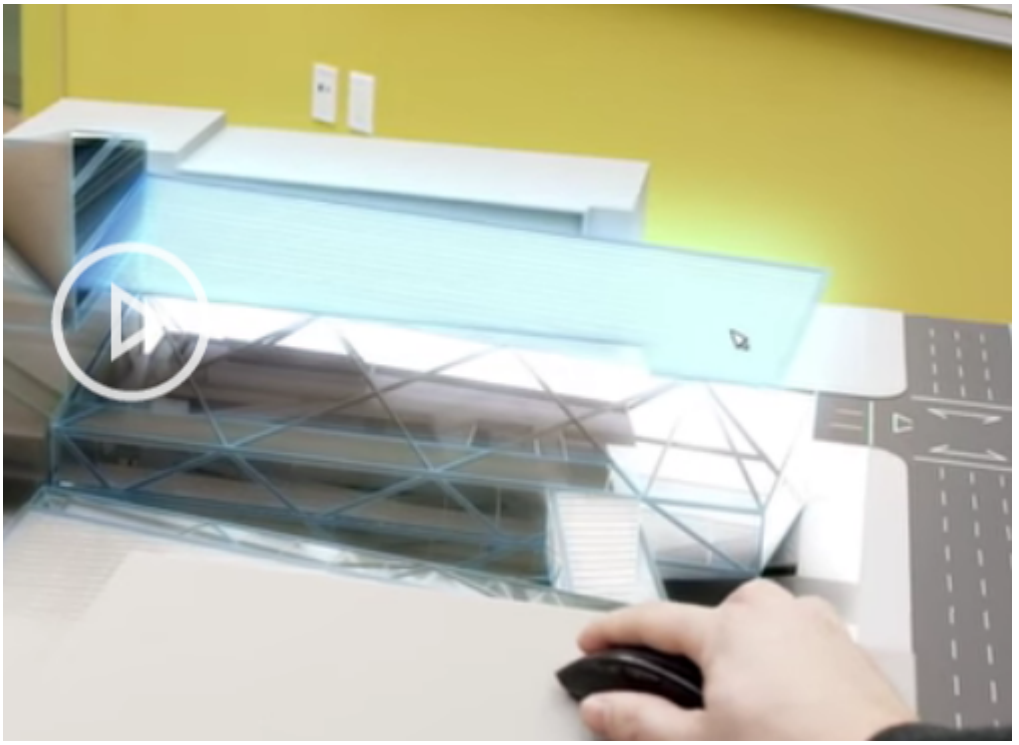
Element 19:

projecting the resulting updated image,

The Hololens updated image in this instance is the roof being raised on ths image via the processor passing those instrutions through to the projector.

https://www.microsoft.com/microsoft-hololens/en-us/commercial

Supporting Evidence:

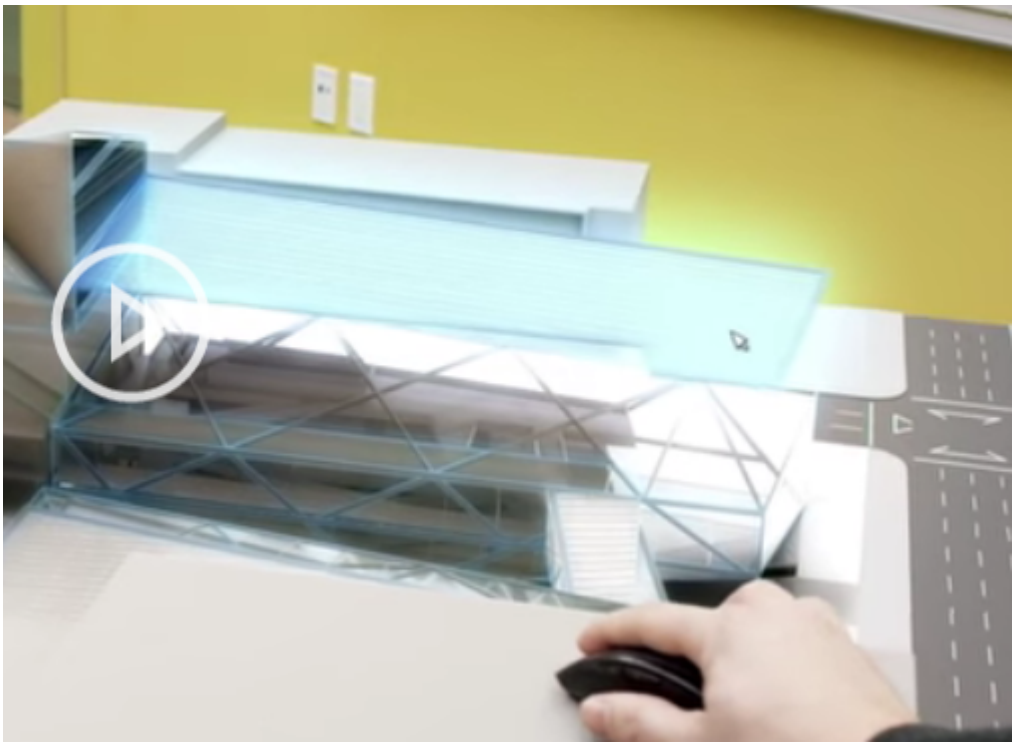


Element 20:

the updated image being equivalent to the initial image

The Hololens created updated image is exactly equivalent to the initial image inclusive of the update.

Supporting Evidence:



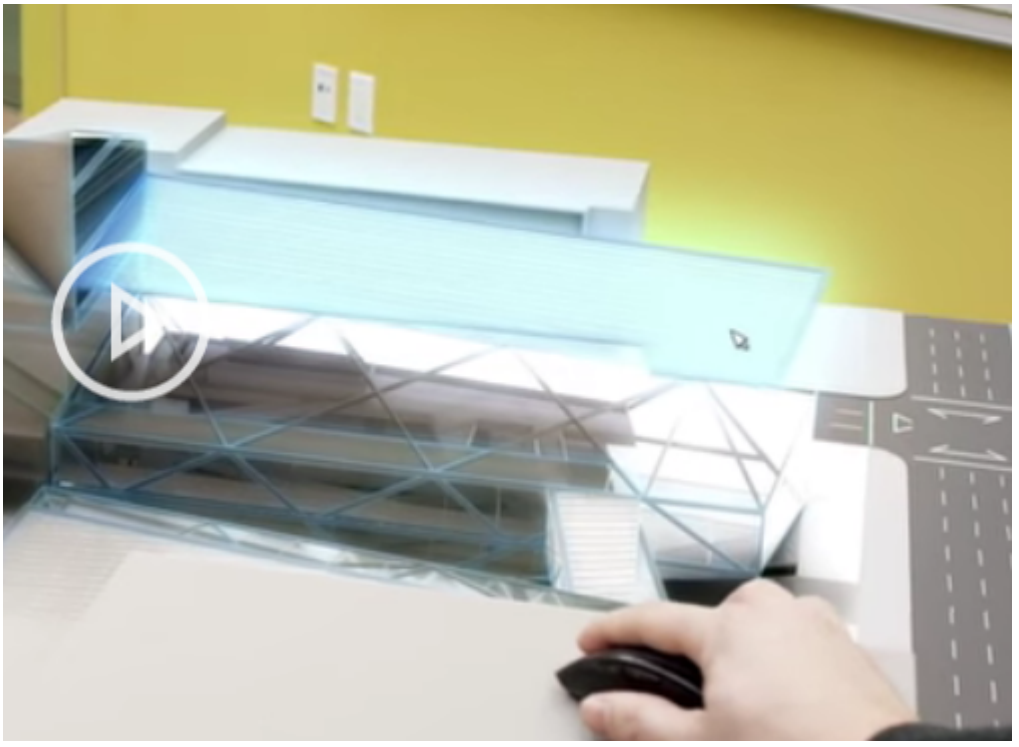
Element 21:

manipulated by a distortion in three dimensional space.

In this instance, that manipulation distortion in the 3D space allowed the user to raise the roof on the image/building design. This is functionality which is applied to all uses of the Hololens.

https://www.microsoft.com/microsoft-hololens/en-us/commercial

Supporting Evidence:



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End of Report

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Rank Here

Evaluation Code: MjZAOThATWljcm9zb2Z0QEhvbG9sZW5zQDc0

We are available to answer follow up questions, provide clarifications and search for additional materials upon request. For public contact - please refer to our profile at ClaimCharts.info. For private contact - please call/email us.

Our hourly rate is: $

Non-Disclosure Agreement:

We understand that for the purpose of assisting you in collecting data, you may disclose to us certain confidential information that you want to keep confidential; therefore, for good and valuable consideration, the receipt and sufficiency of which is hereby acknowledged, we agree as follows:

1. For a period of twelve (36) months from the date we receive your confidential information ("Effective Date"), we will not use or disclose the information, except: (a) in the event that such information (i) comes into the public domain not through us; (ii) is already known to us at the time of disclosure; (iii) is lawfully otherwise learnt or independently developed by us; or (iv) is required to be disclosed by law.

2. Since a breach of our obligations hereunder may result in irreparable harm to you, we agree that upon such breach or threat thereof, you shall be entitled to specific performance and other injunctive relief without posting a bond, and in the event of disclosures required by court order, we shall promptly notify you of the proceeding.

3. This agreement supersedes all prior discussions and writings and constitutes the entire agreement between you and us with respect to the subject matter hereof. This Agreement shall be binding upon the successors and assigns of the respective parties, and may not be modified except by an instrument executed by both parties.

Don Demeter